Predictive and Adaptive Semantic Communication for QoE Optimization in Multi-UAV Networks

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***Abstract*—Deep learning-based semantic communication has demonstrated superior efficiency in wireless image transmission. However, traditional reactive schemes often suffer from outdated Channel State Information (CSI) in highly dynamic multi-UAV environments, leading to severe latency and utility degradation. To address this challenge, we propose the Predictive and Adaptive Semantic Communication (PASC) framework. PASC integrates a GRU-based predictor to anticipate channel evolution, enabling the proactive adjustment of compression rates by dynamically calibrating attention-weight thresholds. Furthermore, a deadline-aware deep reinforcement learning (DRL) algorithm is proposed to jointly assign sub-channels, allocate bandwidth, and adjust power based on predictive states, thereby preventing resource monopolization. Simulation results confirm that PASC achieves a 58.8% improvement in average Quality of Experience (QoE) compared to non-predictive baselines in low-SNR regimes. Crucially, the framework demonstrates formidable robustness to imperfect CSI, strictly bounding end-to-end latency and maintaining high semantic fidelity even in the presence of extreme prediction noise.**

***Index Terms*—Semantic communication, deep reinforcement learning (DRL), channel prediction, adaptive resource allocation, multi-UAV networks.**

I. Introduction

DEEP learning-based semantic communications have emerged as a paradigm shift from traditional bit-oriented transmission to task-oriented information exchange [1]. By extracting and transmitting only essential semantic features [2], this approach significantly enhances transmission efficiency and reliability. As a foundational contribution, Weng et al. designed DeepSC [3], employing a transformer-based approach for efficient text transmission. Building upon these insights, subsequent studies introduced adaptive mechanisms like Universal Transformers [4] and Information Bottleneck (IB) theory for optimized rate-distortion tradeoffs [5]. Furthermore, the practical utility of semantic communications has been extended to multi-UAV networks for image classification, mitigating severe bandwidth and power constraints [6]. Recent works have also explored joint edge computing [7], adversarial training [8], and semantic-aware resource allocation (SARA) [9].

However, the practical implementation of semantic communications in UAV scenarios is hindered by the inherent volatility of wireless channels. Existing schemes often adopt reactive strategies based on outdated CSI [10]. In rapidly fading environments, the adaptation latency causes severe semantic distortion. While sequence models like GRUs capture temporal dependencies [11], existing methods isolate link-level prediction from source coding.

To address these timeliness issues, we propose the Predictive and Adaptive Semantic Communication (PASC) framework. Unlike conventional reactive schemes that suffer from feedback delays, or isolated prediction-aided designs that strictly separate physical-layer forecasting from upper-layer coding, PASC establishes a novel perception-prediction-decision closed-loop. By integrating GRU-based forecasting with a MobileViT engine, PASC proactively calibrates attention thresholds to adjust semantic compression before channel degradation occurs. Simultaneously, a DRL agent leverages these predicted states to jointly optimize sub-channel, bandwidth, and power allocation, fundamentally resolving the adaptation latency in dynamic UAV networks.

The main contributions of this letter are summarized as follows:

- We propose a predictive semantic transmission framework that employs GRUs for proactive channel-state perception, effectively closing the loop between physical-layer forecasting and application-layer coding to mitigate UAV channel fluctuations.
- We introduce an adaptive semantic adjustment mechanism based on MobileViT and attention thresholds, which balances transmission efficiency and semantic fidelity.
- We develop a cross-layer DRL optimization mechanism that incorporates predicted channel gains into the state space, enabling proactive co-optimization of sub-channel, bandwidth, and power allocation.

II. System Model and Problem Formulation

As shown in Fig. 1, this section establishes the PASC framework for multi-UAV networks. By characterizing the

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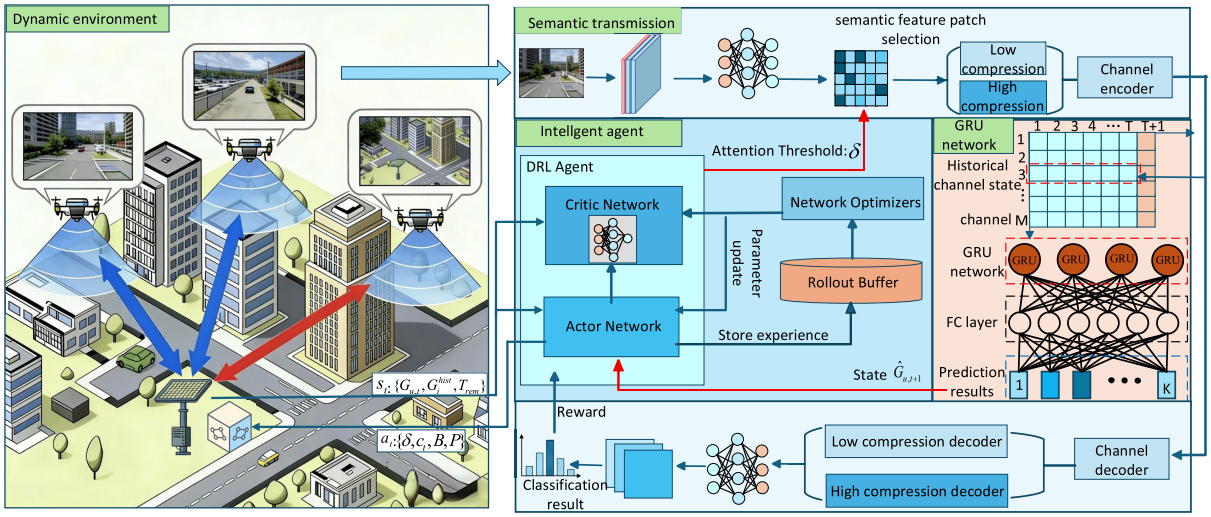


Fig. 1. Framework of the DRL-based joint semantic-resource optimization with GRU channel forecasting.

dynamic environment, we integrate a GRU-based module to proactively perceive channel fluctuations. Finally, a QoE metric is defined that couples semantic utility with end-to-end latency to quantify system performance.

A. Network Model and Channel Prediction

We consider N UAVs performing real-time surveillance along fixed trajectories. Each UAV compresses captured images into semantic symbols for base station (BS) transmission. The wireless link is a block-fading channel [12]. To capture severe UAV mobility-induced Doppler shifts without requiring explicit trajectory optimization, small-scale fading is simulated via the Jakes model, generating a Rayleigh sequence that reflects highly volatile air-to-ground dynamics. The received signal at the BS from the u-th UAV can be expressed as

$$y_{u,t} = \sqrt{P_{u,t}} G_{u,t} s_{u,t} + n_{u,t}, \tag{1}$$

where $P_{u,t}$ is the transmit power, $G_{u,t}$ is the complex channel gain, and $n_{u,t}$ represents the additive white Gaussian noise (AWGN).

To overcome the inherent delay in traditional reactive CSI feedback, we integrate a GRU-based channel prediction module. By exploiting its internal gated mechanism to capture long-term temporal dependencies, the GRU architecture maps the historical trajectory $\mathbf{G}_u^{\text{hist}} = \{G_{u,t-\tau}, \ldots, G_{u,t}\}$ to a predicted gain $\hat{G}_{u,t+1}$ for the forthcoming transmission interval. The predictive perception serves as a "look-ahead" mechanism, allowing the semantic engine to pre-adjust its importance thresholds before the actual channel state degrades or improves.

B. QoE Modeling and Problem Formulation

To evaluate the proposed PASC framework in multi-UAV image classification, we define a task-oriented QoE metric. The QoE model is designed to capture the fundamental trade-off between maximizing Effective Semantic Utility and minimizing the overall latency cost in dynamic aerial environments. Specifically, the formulation integrates the impact of varying channel states and adaptive semantic compression to ensure robust classification performance. Detailed modeling specifications are presented as follows.

1) Effective Semantic Utility: Unlike traditional bit-level metrics, task-oriented semantic fidelity is fundamentally evaluated by downstream execution performance. Therefore, directly mapping classification accuracy to semantic utility $G_{u,t}^{sem}$ rationally quantifies the value of transmitted features. To reflect diminishing marginal utility [13], we employ a generalized Sigmoid-based function:

$$G_{u,t}^{sem} = U(Acc_{u,t}) = \frac{1}{1 + e^{-\kappa \cdot (Acc_{u,t} - \mu)}}, \tag{2}$$

where $\mu = 0.5$ physically represents the mission-critical boundary (below which accuracy resembles random guessing), while $\kappa = 20$ models the rapid utility saturation (the "cliff effect") inherent to AI inferences once essential semantics are reconstructed. Empirically profiled via the MobileViT engine, lowering the attention threshold δ extracts more fine-grained features, enhancing $Acc_{u,t}$ at the expense of proportionally higher bandwidth costs.

2) Comprehensive Latency Cost: The comprehensive latency cost $C_{u,t}^{lat}$, also denoted as the total end-to-end latency $T_{u,t}^{total}$, accounts for the time consumption from image capture at the UAV to the final inference result at the BS. To resolve dimensional inconsistency with semantic utility, the latency cost is normalized against the maximum tolerable delay constraint $T_{\max}$:

$$C_{u,t}^{lat} = \frac{T_{u,t}^{total}}{T_{\max}} = \frac{T_{u,t}^{trans} + T_{u,t}^{comp}}{T_{\max}}, \tag{3}$$

where $T_{u,t}^{comp}$ denotes the total computational overhead, which is further decomposed into the semantic codec latency $T_{u,t}^{sem}$ and the decision-making time $T_{u,t}^{infer}$ of the intelligent agent, calculated as

$$T_{u,t}^{comp} = T_{u,t}^{sem} + T_{u,t}^{infer}. \tag{4}$$

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Specifically, the semantic codec latency $T_{u,t}^{sem} = \alpha k_{u,t} + \beta$ is determined via a linear fitting formula based on the symbol volume $k_{u,t}$. Here, α and β are regression coefficients associated with the specific processing capacity of the hardware. Furthermore, $T_{u,t}^{infer}$ represents the decision-making time, which encompasses the computational overhead of GRU-based channel forecasting and DRL-based resource management, which can be quantified as

$$T_{u,t}^{infer} = T_{u,t}^{pred} + T_{u,t}^{decision}, \tag{5}$$

where $T_{u,t}^{pred}$ and $T_{u,t}^{decision}$ denote the time required for GRU-based channel forecasting and DRL-based joint resource management, respectively. These subtle delays are integrated into the latency constraints to guarantee reliable PASC performance under strict deadlines.

3) Problem Formulation: To maximize long-term cumulative QoE despite highly dynamic environments, we optimize a joint policy $\pi = (\delta, c_t, B_u, P_u)$ at each time slot. The decision-making is subject to several physical constraints. Specifically, the end-to-end latency $T_{u,t}^{\text{total}}$ is capped at $T_{\max}$ to ensure real-time classification performance. Transmit power P_u and aggregate bandwidth B_u are limited by individual hardware budgets $P_{\max}$ and the total spectrum B_{total}, respectively. Furthermore, the attention threshold δ is normalized within $[0, 1]$ for semantic granularity control, while the sub-channel index c_t is selected from an orthogonal set to minimize inter-UAV interference. Accordingly, to navigate this highly non-convex problem, the optimization objective is formulated as the maximization of the expected discounted reward within the DRL framework.:

$$\max_{\{\delta, c_t, B_u, P_u\}} \quad \mathbb{E}\left[\sum_{t=1}^{T}\sum_{u=1}^{N} QoE_u(t)\right] \tag{6}$$

$$\text{s.t.} \quad T_{u,t}^{\text{total}} \le T_{\max}, \tag{6a}$$

$$0 \le P_u \le P_{\max}, \tag{6b}$$

$$\sum_{u=1}^{N} B_u \le B_{\text{total}}, \tag{6c}$$

$$0 \le \delta \le 1, \tag{6d}$$

$$c_t \in \{1, \ldots, M\}, \tag{6e}$$

where (6a), (6b), and (6c) denote the latency constraints, transmit power, and total bandwidth, respectively. The constraint (6d) limits the range of the attention threshold for adaptive feature selection, while (6e) defines the feasible set of sub-channel indices available for UAV transmission.

III. The Proposed Framework

This section details the implementation of the proposed PASC framework. Its core mechanism involves a DRL agent that proactively optimizes semantic granularity and resource allocation based on predicted channel states. Through pre-established mapping relationships, the framework achieves fine-grained adaptation to maintain real-time mission reliability.

A. Predictive Semantic Encoding Engine

As shown in Fig. 1, the proposed state-action-reward (S-A-R) closed loop couples GRU prediction, semantic encoding, and DRL resource allocation. The MobileViT backbone [13] extracts feature patches and applies self-attention to assign normalized weights $w \in [0, 1]$, generating a spatial importance map $\mathbf{W}$. To minimize latency, the optimal attention threshold δ dynamically adapts to the GRU-predicted channel gain $\hat{G}_{u,t+1}$. Analytically, the allowed transmission duration is bounded by $T_{u,t}^{trans} \le T_{\max} - T_{u,t}^{comp}$. Since a fading channel (lower $\hat{G}_{u,t+1}$) reduces the transmission rate, averting timeouts strictly requires raising δ to decrease the symbol volume K, yielding:

$$\delta \propto \frac{1}{\hat{G}_{u,t+1}}. \tag{7}$$

To minimize real-time computational overhead, an offline-profiled mapping $K = \mathcal{M}(\delta)$ enables robust $\mathcal{O}(1)$ online retrieval of the required symbol volume. Once δ is determined, feature patches are partitioned into critical and non-essential subsets. The total transmitted symbol volume is then given by:

$$K = N_{\text{imp}}(\delta) \cdot k_{\text{low}} + N_{\text{non}}(\delta) \cdot k_{\text{high}}, \tag{8}$$

where $N_{\text{imp}}(\delta)$ and $N_{\text{non}}(\delta)$ denote the cardinality of important ($w \ge \delta$) and non-essential patches, which are compressed using k_{low} and k_{high} symbols/patch, respectively.

B. Predictive Resource Allocation and QoE Co-Optimization

We optimize the cumulative QoE using Proximal Policy Optimization (PPO) augmented with Generalized Advantage Estimation (GAE). The agent observes a "look-ahead" state space $\mathbf{s}_t = \{\hat{G}_{u,t+1}, G_{u,t}, \gamma_t, T_{rem}\}$, integrating GRU-predicted gains and mission deadlines to mitigate reactive adaptation latency fundamentally. Based on $\mathbf{s}_t$, the policy network outputs a joint action $\mathbf{a}_t = \{\boldsymbol{\delta}, B_u, P_u, c_t\}$.

To theoretically ensure stable convergence under severe UAV mobility, PPO optimizes a clipped surrogate objective, which is mathematically defined as

$$L^{CLIP} = \hat{\mathbb{E}}_t\left[\min\left(r_t\hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{9}$$

where r_t is the policy ratio and $\epsilon = 0.2$ bounds the update. Crucially, the advantage $\hat{A}_t$ is estimated via GAE, which exponentially discounts temporal difference errors to smooth volatile reward fluctuations induced by highly dynamic UAV channels. This effectively prevents catastrophic performance collapse.

The reward $R_{u,t}$ incorporates a stringent penalty $-\beta_{\text{timeout}}$. To fundamentally align the DRL optimization with hard latency constraints, the algorithm leverages the discount factor to compel the agent to prioritize immediate deadline satisfaction over distant returns. Consequently, if $T_{u,t}^{\text{total}} \le T_{\max}$, it yields the instantaneous QoE; otherwise, utility is strictly nullified:

$$R_{u,t} = \begin{cases} w_1 G_{u,t}^{sem} - w_2 C_{u,t}^{lat}, & \text{if } T_{u,t}^{total} \le T_{max}. \\ -\beta_{timeout}, & \text{if } T_{u,t}^{total} > T_{max}. \end{cases} \tag{10}$$

As summarized in Algorithm 1, the PASC framework achieves proactive cross-layer optimization by integrating offline pre-trained models with an online DRL execution phase. At each time slot, the agent observes the predictive environment state to dynamically sample a joint action a_t, which seamlessly aligns the semantic granularity with physical resource allocation. Crucially, the algorithm explicitly

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Algorithm 1 The PPO-Based Optimization of Prediction-Driven Semantic Communication

Input: Pre-trained MobileViT & GRU, Mapping $\mathcal{M}$, $T_{\max}$.

for slot $t = 1, \ldots, T$ **do**
  Observe state $s_t = \{\hat{G}_{u,t+1}, G_{u,t}, \gamma_t, T_{rem}\}$
  Sample joint action $a_t = \{\delta, B_u, P_u, c_t\} \sim \pi_\theta(\cdot|s_t)$
  Obtain $K \leftarrow \mathcal{M}(\delta)$ and compute delay $T_{u,t}^{\text{total}}$
  Set $R_{u,t} = QoE_u(t)$ if $T_{u,t}^{\text{total}} \le T_{\max}$ else 0
  Update policy π_θ using reward $R_{u,t}$
end for

Output: Optimized policy π_θ^*.

accounts for end-to-end temporal costs; any decision that violates the stringent latency constraint $T_{\max}$ is strictly penalized with zero utility. Through continuous environmental interaction and policy updates, the framework ultimately outputs an optimized policy π_θ^* capable of maximizing long-term communication QoE and satisfying real-time mission requirements in highly dynamic UAV networks.

IV. Simulation Results

In this section, we provide a comprehensive evaluation of the proposed framework. The simulations are conducted in a dynamic UAV environment with Jakes fading channels, utilizing the CIFAR-100 dataset for the image classification task.

A. Simulation Settings

For the simulation setup, the multi-UAV network comprises $N = 3$ UAVs and 10 sub-channels over a total bandwidth of $B_{\max} = 1.5$ MHz, under a maximum power of $P_{\max} = 1$ W and a hard delay constraint of $T_{\max} = 150$ ms. Task performance is governed by the Sigmoid-based utility model with a threshold of $\mu = 0.5$ and a sensitivity factor of $k = 20$. The predictive semantic engine integrates a pre-trained MobileViT backbone with a 3-layer, 256-unit GRU predictor, where the Conv2D layer operates with $K = S = 2, F = 128$, and the core block features a depth of 2 with 4 attention heads. Accordingly, the PPO agent employs $[256, 128]$ multi-layer perceptrons for policy and value networks with a clipping range of $\epsilon = 0.2$.

B. Performance Analysis

Fig. 2 compares the convergence behaviors of different strategies. Integrating GRU-based channel prediction into the DRL state space accelerates convergence and yields a higher steady-state reward. Foreseeing channel trends avoids drastic reward fluctuations from deep fading, ensuring a smoother trajectory than the baseline.

Fig. 3(a) shows that PASC consistently outperforms baselines in QoE, particularly in low-SNR regimes. Unlike reactive schemes suffering deadline violations due to stale CSI, PASC dynamically tunes K via predictions, avoiding over-optimistic decisions that nullify utility. By aligning semantic granularity with predicted capacity, the framework strictly satisfies delay thresholds, ensuring reliable communication.

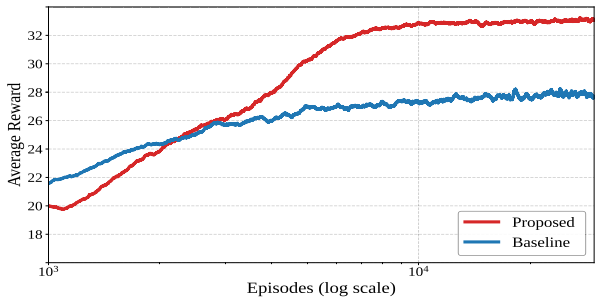


Fig. 2. Training convergence comparison.

TABLE I
Quantitative Comparison of Prediction Fidelity

Metric	Baseline (LSTM)	Proposed (GRU)	Improvement
RMSE (dB)	5.659	1.062	81.2% ↓
MAE (dB)	4.382	0.505	88.5% ↓

Fig. 3(b) evaluates the Effective Semantic Utility over varying channel gains. Under the designated Sigmoid-based utility configuration, PASC maintains high utility even under extreme fading. By predicting channel degradation, the framework prioritizes critical task-relevant features to maximize semantic value per bit, whereas reactive schemes suffer rapid performance collapse. Consequently, this proactive adaptation ensures graceful degradation, effectively mitigating the conventional cliff effect.

Fig. 3(c) evaluates the total end-to-end delay T_{total}. Despite a minor forecasting overhead, PASC yields the lowest latency across all SNR regimes. By capturing transient channel dips, the system consistently satisfies the 150 ms hard deadline, avoiding costly timeout penalties. Crucially, the underlying deadline-driven reward design guarantees a stable access rate and resource fairness, preventing bandwidth monopolization at the expense of any individual user.

Fig. 4(a) isolates module impacts by comparing PASC against variants without GRU prediction or the adaptive threshold. The results reveal a strong interdependency: removing either component severely degrades QoE due to increased timeout penalties or compromised semantic fidelity, respectively. The necessity of the joint paradigm is thus confirmed.

Fig. 4(b) illustrates the GRU module's high-fidelity "look-ahead" capability. Quantitatively, Table I demonstrates that GRU reduces RMSE and MAE by over 81% compared to the LSTM baseline. Fundamentally, GRU's streamlined gating mechanism effectively mitigates the vanishing gradient problem inherent to highly volatile Jakes fading channels, while its simplified architecture substantially lowers decision-making latency. This precise perception enables pre-emptive adjustment of semantic parameters.

Fig. 4(c) evaluates the robustness of PASC against imperfect CSI. PASC maintains a stable QoE advantage over baselines, exhibiting only graceful degradation across varying prediction error variances. The system demonstrates high resilience against forecasting inaccuracies inherent to dynamic aerial environments.

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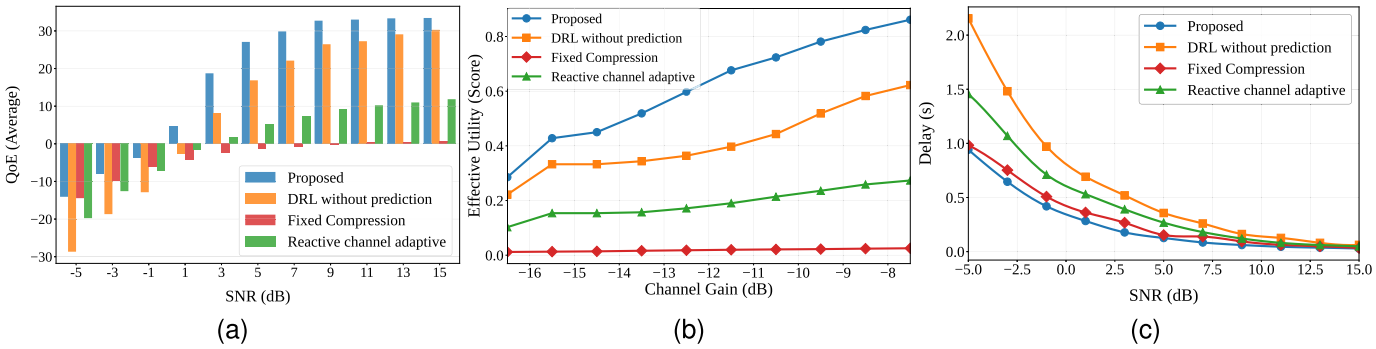


Fig. 3. Performance analysis: (a) Average QoE vs. SNR, (b) semantic utility vs. channel gain, and (c) end-to-end delay vs. SNR.

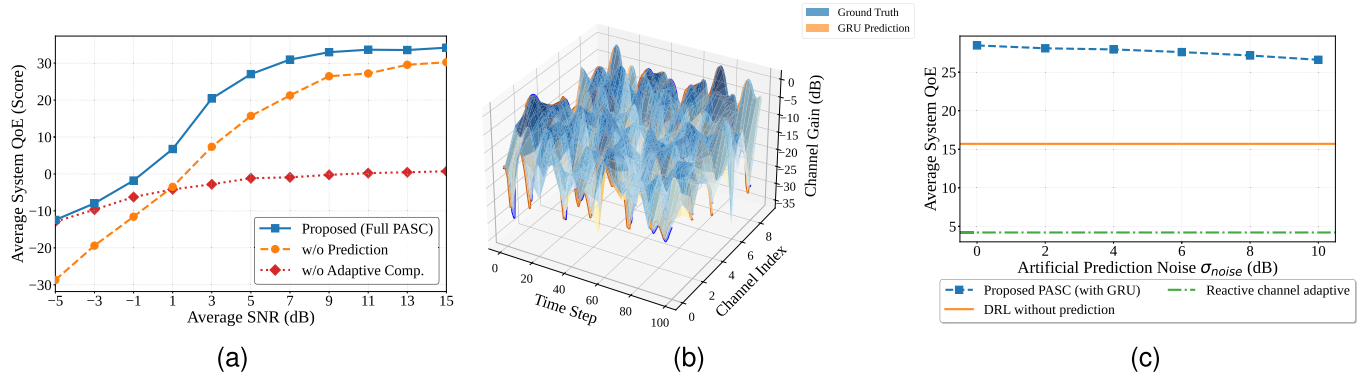


Fig. 4. In-depth system analysis: (a) ablation study validating module synergies, (b) high-fidelity GRU channel prediction, and (c) system robustness against imperfect CSI.

V. Conclusion

In this letter, the Predictive and Adaptive Semantic Communication framework was proposed to maximize QoE in multi-UAV networks. By synergistically integrating GRU-based deep channel forecasting with a MobileViT-driven semantic engine, PASC established a proactive "look-ahead" mechanism to dynamically adjust semantic compression thresholds. Governed by a deadline-aware DRL strategy, the framework jointly optimized sub-channel assignment, transmission power, and bandwidth allocation, thereby preventing resource monopolization, ensuring equitable user access, and balancing semantic utility against temporal costs. Extensive simulations demonstrated that PASC significantly outperformed state-of-the-art reactive baselines, delivering a 58.8% average QoE enhancement. Furthermore, the developed system exhibited remarkable robustness against severe fading and imperfect CSI, strictly satisfying hard latency constraints while maximizing semantic value in low-SNR regimes. This framework paves the way for semantic-aware aerial networks under strict latency and reliability constraints.

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